Validation protocol reverses machine-learning rankings in tokamak confinement scaling

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4 September 2026

Code and data: `https://github.com/lsalcedo100/FusionFlux`
Archived at doi:10.5281/zenodo.22215142

Abstract

Confinement scaling laws set the size a tokamak must reach to ignite, and are fitted across the devices already built. On the ITPA global H-mode database (HDB5, standard analysis set STD5; 6228 slices from 4471 discharges across 18 tokamaks), under cross-validation grouped by discharge, a random forest reduces log error by 29% relative to a blind refitted power law and by 36% relative to the published IPB98(y,2) reference. That margin does not survive the question a scaling law exists to answer. Under leave-one-tokamak-out the ordering of the three primary fitted models reverses exactly: the random forest is worse than the same log-linear power law on all 13 held-out database labels and all 11 physical devices (sign test $p = 9.8 \times 10^{-4}$). The reversal survives a hyperparameter search nested inside every training fold. The forest's error tracks the held-out machine's distance from the training data ($\rho = +0.85$) where the power law's does not ($\rho = -0.06$), and it cannot predict outside its training target range. At a size-ordered cut reproducing ITER's 1.82× jump in major radius, the tree ensembles score closer to a constant predictor than to the power law, and split-conformal intervals calibrated to 90% fall to 3% row-level coverage at unchanged width. Two repairs work, both adding long-range structure rather than flexibility: a Connor–Taylor constraint surface, and a Gaussian process whose linear component keeps trending. Across the families tested here, extrapolation failure tracks long-range saturation rather than flexibility alone. For ITER the surviving and failing models differ by a factor of 8.3.

1 Introduction

A tokamak ignites when the alpha heating from its own fusion reactions covers everything the plasma loses. The energy confinement time τ_E sets that loss rate, which makes it the quantity that decides how large a next-step device has to be, and no first-principles calculation predicts it to the accuracy a design decision needs. The field's working substitute is an empirical scaling law: a power law in the engineering parameters,

$$\tau_E = C\, I_p^{a_1} B_t^{a_2} \bar{n}_e^{a_3} P^{a_4} R^{a_5} \epsilon^{a_6} \kappa^{a_7} M^{a_8}, \tag{1}$$

fitted by least squares across the discharges of every machine that has run. In log space Eq. (1) is exactly a linear model, so fitting one is ordinary least squares on a log design matrix. The reference fit is IPB98(y,2) [1], a regression over the ITPA multi-machine database whose exponents were used to size ITER; that database is still maintained and republished [2].

What such a law is used for is an extrapolation, and a large one. ITER's major radius is 1.82 times the largest machine in the database, and SPARC [10] is designed for a toroidal field three times anything in it. The law is asked for a number at a point no row occupies, and that is the regime in which a fitted regression carries no guarantee at all.

Replacing Eq. (1) with a flexible regressor is an obvious move, and machine learning has been productive elsewhere in this field: disruption prediction [12], transfer of a disruption predictor to a device that supplied almost none of its training data [13], and real-time magnetic control [14]. Confinement scaling itself has been revisited with symbolic regression, which abandons the power-law form rather than fitting it [15], and with a neural residual learned on top of a frozen power law [16]. Measured conventionally the flexible models win. On the rows used here the random forest scores 0.128 in log-RMSE against 0.181 for a power law refitted blind on the same rows and features, a margin of 29%, and against 0.199 for the published IPB98(y,2) law, a margin of 36%. The two comparators are not interchangeable and both are quoted throughout: IPB98(y,2) was fitted on this database including whichever machine is held out, so it is a reference point rather than a blind baseline, and the blind refit is the like-for-like comparison. The reversal

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below is stated against the blind refit. The repository accompanying this paper quotes 41% for the same model, scored on the ten-column matrix of Sec. 3, which adds IPB98(y,2)'s own prediction to the nine engineering features. Every number below uses the nine.

Related work on this database. Four recent studies apply machine learning to the same ITPA data and are the closest prior work. Nam and Seo [24] take cross-machine transfer as their subject: they align features to correct the covariate shift between devices and use deep ensembles for uncertainty, validating on conditions the model has not seen, including JET-ILW and ITER. Gao *et al.* [25] apply variational and sampling-based Bayesian neural networks to STD5 together with individual JET, DIII-D and ASDEX Upgrade data, reporting improved accuracy alongside posterior uncertainty. Xiong *et al.* [26] tune random forests and two neural architectures by Bayesian optimisation on STD5 and report R^2 up to 0.990, with a feature-importance analysis separating spherical from conventional tokamaks. Wang *et al.* [16] freeze a power law and learn its residual, reporting improved stability under shifts defined by parameter ranges.

This paper asks a different question, and the difference is the protocol rather than the model. None of these works is the target of a criticism here; what any of them reports is what its split measures. That is the point. Where a score on this database is computed without holding out a whole device, it describes discharges from machines the model has already seen, and Sec. 5 shows that the ranking such a split produces can invert completely once the held-out unit becomes a machine. We do not propose a better predictor. We ask whether the validation that selects one answers the question a scaling law exists for, measure what the honest answer costs, identify the property of a model that decides it, and test the repair that physics rather than flexibility supplies.

That measurement is the wrong one, for a reason about the data rather than about the models. Cross-validation on this database holds out *discharges*, and every machine in a held-out fold also appears in the training fold, so a held-out row has hundreds of near neighbours recorded on the same device at nearby settings. What the score quantifies is how much of JET is predictable from the rest of JET. That a validation split should match the generalisation being claimed, and that grouped observations flatter a model when it does not, is well understood in fields whose data arrives in clusters [17]; it is not how a candidate confinement model is usually scored. A scaling law is not for the machines that exist.

This paper measures what the deployment-matched split costs. We escalate what is held out in three steps, a discharge, then a whole machine, then an entire size range, and score the same models under each. The results are: (i) under leave-one-tokamak-out the ranking of the three primary fitted models is the exact reverse of their ranking under grouped cross-validation, and the best cross-validated model loses to a log-linear power law on all 13 held-out database labels and all 11 physical devices; (ii) three measurements of why, covering extrapolation distance, the training-target bound of a random forest, and what polynomial flexibility costs in the tail; (iii) a size-ordered split inside the database that reproduces ITER's extrapolation factor, at which conformal intervals calibrated to 90% cover 3% of rows without becoming any wider; (iv) the Connor–Taylor dimensional constraints, imposed as linear equality constraints on the exponents, giving the lowest observed error of any fit tested here at that cut with nothing to tune, on a surface selected after those scores were seen; (v) a Gaussian-process experiment that separates flexibility from long-range saturation, holding the nonlinear component's specification fixed and varying only the mean-function specification, and corrects the diagnosis the earlier sections leave open; (vi) replication on rows the standard analysis set does not contain, in a second confinement regime, and on two scaling laws from biology, with a predictor-count ladder that separates the extrapolation failure, which persists at every predictor count tested, from the ranking inversion, which appears only once the flexible model wins the interpolation split; and (vii) a locked forecast for SPARC, JT-60SA and ITER. Section 3 begins with a property of the design matrix itself: it is rank deficient by two, and one of the two dependencies is that a published scaling law added as a model feature contributes nothing a log-linear fit can use.

2 Data and method

Data. The ITPA Global H-mode Confinement Database, standard analysis set STD5 v5.2.3 [2], obtained from OSF. After requiring finiteness and positivity, 6228 rows from 4471 discharges across 18 tokamaks remain. The target is the thermal confinement time `TAUTH`. No synthetic data appears in any result. The analysed file is pinned by SHA-256 and verified on load, so every number below refers to one specific set of bytes rather than to whatever the host currently serves.

Features and models. Nine log engineering features: plasma current, toroidal field, line-averaged density, loss power, major radius, elongation, inverse aspect ratio, effective mass and minor radius. The analytic IPB98 prior is *excluded* as a feature throughout the extrapolation arms, because its exponents were fitted on this same database including whichever machine is held out; retaining it leaks the answer into every fold. The model set is a ridge regression on the log features (i.e. a fitted power law), a random forest [20] (300 trees), a histogram gradient booster, polynomial ridge controls of degree 2 and 3, and a mean baseline. IPB98(y,2) evaluated analytically is reported as a physics reference, and is marked throughout as *not blind*.

What is and is not tuned. Every model predicts $\log \tau$ and is exponentiated to τ before scoring. None of the models in Table 1 is hyperparameter-tuned, and "library defaults" is not a specification, so the consequential settings are given here in full. The power law is `StandardScaler` followed by `Ridge` with $\alpha = 1$ and the SVD solver, chosen before any held-out score was computed and never varied. The random forest is `RandomForestRegressor` with 300 trees, `criterion="squared_error"`, unlimited depth, `min_samples_split` = 2, `min_samples_leaf` = 1, `max_features` = 1.0 and bootstrap resampling. The gradient booster is `HistGradientBoostingRegressor` with squared-error loss, learning rate 0.1, at most 100 boosting iterations, 31 leaves, no depth limit, `min_samples_leaf` = 20, no L_2 penalty and 255 bins. Its `early_stopping="auto"` resolves to *off* at every sample size here, since that setting enables early stopping only above 10 000 rows and no training fold reaches that. The polynomial controls are `PolynomialFeatures` of degree 2 and 3 without a bias column, then the same scaler and ridge. Neither tree ensemble is standardised, which is immaterial to both.

Every estimator carries the seed 42, and the forest is fitted in parallel but predicts single-threaded so that reruns are bit-identical. Standardisation sits inside the pipeline, so it is refitted on the training rows of every fold and never sees a held-out row. No within-model hyperparameter search is performed on any reported cross-validation score, so nested tuning is not required for those numbers, and Sec. 5.2 reports what a nested search does when one is run anyway. Comparing several prespecified model families is still a selection of a kind, and two later choices go further: the constraint rung of Sec. 9 and the damping of Sec. 7 were both picked after seeing held-out scores, and both are flagged where they appear. The Gaussian processes of Sec. 13 do estimate kernel hyperparameters, by marginal likelihood on training rows alone.

Splits. (i) Grouped cross-validation by discharge: `GroupKFold` with five folds, grouped on the discharge identifier, so every slice from one discharge falls in the same fold. It is constructed with `shuffle=False`, its default, so the partition is deterministic and no seed enters it. (ii) Leave-one-tokamak-out over the 13 machines with at least 30 rows: each is held out in turn and the model is trained on the other 12. (iii) A size-ordered cut: machines are ordered by median major radius, the model is trained below a cut and scored on every machine above it, and the reported cut is the one whose size ratio is closest in log to the ratio between this database and ITER. That rung has a ratio of 1.823, trains on 14 machines and scores 2730 held-out rows, and is called the ITER-size-matched cut throughout. The match is to ITER's major-radius ratio and to nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient.

The score is the root-mean-square error in $\log \tau$, $\mathrm{RMSE}_{\log \tau} = \sqrt{\overline{(\log \hat{\tau} - \log \tau)^2}}$, written log-RMSE here and in the figures. It is the plain log residual, not the $\log(1 + y)$ quantity usually called RMSLE; confinement times are strictly positive, so no offset is needed, and the name is avoided here to keep the two distinct. Intervals are percentile bootstraps resampling whole *machines*, since the claim concerns behaviour on an unseen machine: 2000 resamples drawn with replacement over the 13 machines, seeded at 20240617, reported as the 2.5th and 97.5th percentiles. Paired intervals resample the per-machine difference rather than the two error levels, which removes the shared machine difficulty.

Preprocessing. Rows are dropped only for non-finiteness or non-positivity in a used column, which is what leaves 6228 of the delivered rows; no outlier rule, no winsorising and no reweighting is applied. Features are natural logs of the raw engineering columns, taken after cleaning. Minor radius is reconstructed as $a = \epsilon R$ at this stage, which is the origin of one of the two exact collinearities of Sec. 3. Wall variants (JET-C and JET-ILW, and the two ASDEX Upgrade entries) are separate labels in the database and are kept separate except where the text says devices are collapsed to 11.

Software. Python 3.12.14, scikit-learn 1.7.2, NumPy 2.2.6, SciPy 1.18.1 and pandas 2.3.3. The version of every library is pinned in the repository's `constraints.txt`, and the defaults quoted above are the ones those versions carry; "defaults" in a later release may differ, which is why they are written out rather than referenced.

3 The design matrix is rank deficient by two

Three column counts appear in this paper and they are not the same set. The models see nine log engineering features. Adding the analytic IPB98 prior to those nine gives the ten-column matrix examined here. The refit of Eq. (1) in Sec. 4 drops both redundant columns and uses nine, eight independent variables plus an intercept, which is why its condition number is a healthy 10.7.

Standardised, the ten-feature matrix has rank 8, with the last two singular values at 4.9×10^{-14} and 2.4×10^{-14}. The models keep the redundant nine-column representation, which is free for least squares but not obviously so for ridge, since a penalty is applied in the coordinates it is handed. Minor radius is a parameter a practitioner building this model would naturally supply, so nine columns is the set a reader would arrive at unprompted; promoting the eight-column variant because it turns out cleaner would be a selection on held-out scores of the kind Sec. 9 flags in its own result. The eight-column fit is therefore reported as a control rather than as the headline. Dropping `log_a_m` and refitting on eight independent columns leaves the log-linear scores unchanged to four decimals under all three splits, 0.181, 0.214 and 0.278, moves the tree ensembles by at most 0.035, and changes no ordering anywhere.

The estimator is not load-bearing either. Ordinary least squares on those same eight independent columns, carrying no penalty at all, scores 0.1812, 0.2141 and 0.2790 against the reported ridge's 0.1812, 0.2141 and 0.2778: identical to six decimals under cross-validation, apart by 0.0001 under leave-one-machine-out, and by 0.0012 at the size cut. At $\alpha = 1$ on standardised columns with thousands of rows the penalty does almost nothing, so the ridge here is a numerically stable spelling of the least-squares power law rather than a regularised alternative to it. Neither the choice of estimator nor the redundant column carries any of the results below. Projecting candidate vectors onto the null space confirms two exact dependencies at relative residuals of order 10^{-16}: minor radius is derived as $a = \epsilon R$ during cleaning, and the IPB98 prior is a fixed log-linear combination of the other eight features.

The second is the substantive one. Adding a published physics scaling as a model feature feels like adding knowledge; in log space it provably is not, and it contributes exactly nothing to a log-linear fit. Nothing failed loudly because `scipy.linalg.lstsq` inverts through the SVD pseudoinverse and silently returns the minimum-norm member of a two-parameter family of coefficient vectors that fit identically well.

We fell into two methodological traps first. The null space is a *subspace*, so when the deficiency exceeds one the basis vectors printed by the SVD generally do not resemble the dependency sought even when it is exactly right; the correct test is $\|v - N^{\top}Nv\|$. And standardisation rescales the null space, so a raw dependency must be mapped into standardised coordinates before comparison. Omitting this gives residuals of 0.41 and 0.96 for dependencies that are exactly true.

4 Refitting, and where the disagreement lives

Refitting Eq. (1) on all STD5 rows (design matrix 6228×9, condition number 10.7) gives I_p 1.08 against the published 0.93 and R 1.58 against 1.97, while P and B_t return almost exactly. This disagreement is expected: IPB98 was fitted to a selected ITER-relevant subset, whereas this fit applies no selection criteria and includes the spherical tokamaks. A refit on a different population is a different regression.

The normal equations, QR and the SVD pseudoinverse were each implemented from scratch, including the triangular substitutions, and agree to 7.6×10^{-13}. Cholesky on the normal equations is five times faster and three orders of magnitude less accurate, which is the expected trade: it forms $X^{\top}X$ and so works at the *square* of the condition number. At $\kappa = 10.7$ that costs nothing real, and it is the only one of the three that fails loudly on a rank-deficient matrix.

Where the two differ matters more than by how much. Decomposing the difference between the exponent vectors along the singular directions of the design matrix, 77% of it lies in the single weakest direction, which carries 0.3% of the matrix's variance, while the three strongest directions carry 82% of the variance and account for 0.75% of the disagreement. That weak direction is plasma current traded against machine size, which is weak for a structural reason: tokamaks are not designed to vary the two independently, so the existing multi-machine design leaves that direction weakly identified. More shots at the operating points these machines already run would not separate it; shots deliberately placed to break the correlation could.

Table 1: The headline comparison under the conventional reporting choices: the same models and the same nine blind features, scored one way per split. The two columns differ in more than the held-out unit, since they also differ in row population and in how per-row scores are aggregated, and the 13 labels are not 13 independent devices. Sec. 5.2 matches the population and the aggregation and repeats the comparison over physical devices; the inversion survives all of it. IPB98 is a reference, not a blind competitor.

Model	CV	LOMO	Ratio	CV rank	LOMO rank
random forest	0.128	0.465	3.6×	1	5
hist gradient boosting	0.130	0.359	2.8×	2	4
ridge, log-cubic	0.148	0.849	5.7×	3	6
ridge, log-quadratic	0.158	0.300	1.9×	4	3
ridge, log-linear	0.181	0.214	1.2×	5	2
IPB98(y,2), analytic	0.199	0.188	1.0×	6	1
mean baseline	0.869	0.994	1.1×	7	7

5 The ranking inversion

Table 1 is the central result. Among the three primary models, a power law, a random forest and a gradient booster, the order under one split is the exact reverse of the order under the other. The random forest is the best model in the study by cross-validation and the worst of the three on a machine it has not seen. Both columns compare the same models on the same rows and features, so the reversal is a like-for-like statement: the forest is 29% better than the blind power law in the CV column and 117% worse than it in the LOMO column. The analytic IPB98(y,2) row is the published reference and is marked not blind, since it was fitted on this database with every machine included. The two polynomial rungs are controls, scored in the same table but outside the ranking claim, and including them breaks the inversion rather than extending it: the log-cubic rung is third of the five blind models under cross-validation and last of them on a held-out machine.

Because the machines differ enormously in difficulty, the marginal bootstrap intervals overlap and are the wrong test; resampling the *paired* per-machine difference removes that shared difficulty. The random forest is worse than the log-linear power law on **13 of 13** database labels, mean gap +0.251, 95% interval $[+0.157, +0.342]$. The gradient booster is worse on 12 of 13. Section 5.2 repeats this over the 11 physical devices, where the labels are no longer two eras of one machine counted twice, and the gap widens to +0.315 with an interval of $[+0.221, +0.408]$.

5.1 Three diagnostics of the failure

Distance. Ranking each model's per-machine error against the Mahalanobis distance of that machine's mean log-feature vector from the training mean gives $\rho = +0.85$ for the random forest and +0.54 for the gradient booster, against $\rho = -0.06$ for the log-linear power law. The random forest's error rises strongly with extrapolation distance and the gradient booster's trends upward more weakly. The power law's does not: its error is uncorrelated with how far the machine sits from anything it saw.

Thirteen points is few enough that each of those numbers needs an uncertainty. Permuting the distances 20000 times gives $p = 0.0007$ for the forest, $p = 0.060$ for the booster and $p = 0.85$ for the power law, and dropping each machine in turn moves the forest's ρ only within $[+0.81, +0.88]$, so no single extreme device carries it. The booster's +0.54 does not clear a conventional threshold and we read it as suggestive rather than established. One control changes how the rest should be read: the *mean baseline*, which fits nothing, also correlates with distance, at +0.56. Distance predicts how hard a machine is in general. What is unusual is not that the trees degrade with it but that the power law does not.

On the four most distant machines (PBX-M, NSTX, C-Mod, JFT-2M, all small, one of them spherical) the forest's error is 2.7 to 4.8× the power law's; on the four nearest it is within 1.7×.

A hard bound. A random forest predicts by averaging leaf values, and every leaf value is itself a mean of training targets, so every prediction it can make lies in $[\min y_{\text{train}}, \max y_{\text{train}}]$ whatever the features say. With JET held out, 48% of its rows exceed the maximum confinement time in the remaining 12 machines and its best shot is 3.7× above anything any tree in the forest can output. More rows would close this only if they carried larger targets: within the same target range no amount of additional data, and no tuning, moves a ceiling that is a property of the functional form.

A gradient booster carries no such guarantee. It sums tree outputs onto an initial estimate rather than averaging targets, and nothing in that construction confines the sum to the training range, so the

Table 2: Per-machine log-RMSE under leave-one-tokamak-out, ordered by Mahalanobis distance of the held-out machine's mean log-feature vector from the training mean (computed through the pseudo-inverse, since that covariance is singular by Sec. 3). The trees degrade with distance; the power law does not.

Held out	rows	dist.	IPB98	ridge	hist GB	forest
D3D	388	1.1	0.252	0.251	0.246	0.406
AUG	1377	1.2	0.197	0.216	0.281	0.279
AUGW	767	1.6	0.218	0.207	0.212	0.219
JETILW	866	1.7	0.257	0.216	0.244	0.318
JET	1762	2.2	0.148	0.198	0.487	0.478
JT60U	100	2.5	0.275	0.285	0.307	0.291
PDX	97	4.0	0.227	0.233	0.323	0.407
ASDEX	431	5.2	0.131	0.194	0.207	0.507
MAST	39	5.4	0.136	0.154	0.317	0.581
JFT2M	69	5.4	0.095	0.094	0.234	0.450
CMOD	45	6.9	0.119	0.173	0.569	0.521
NSTX	185	7.8	0.222	0.289	0.686	0.857
PBXM	59	10.2	0.172	0.274	0.559	0.727

Table 3: The three contenders under both row populations, both aggregations, and both definitions of a machine. The inversion is the pattern of the random forest leading under cross-validation and trailing the power law under leave-one-out; it holds in all eight combinations.

	pooled over rows		each unit equally	
Split	forest	ridge	forest	ridge
CV, all 18 machines	0.128	0.181	0.149	0.238
CV, the 13 scored machines	0.123	0.177	0.124	0.187
leave-one-label-out (13)	0.410	0.214	0.465	0.214
leave-one-device-out (11)	0.551	0.200	0.527	0.212

argument above does not transfer to it. What we can report is that it never uses the freedom: over the 13 leave-one-out folds and the size cut of Sec. 6, its largest prediction stays below $\max y_{\text{train}}$ on every one, coming closest with JET held out at 0.066 in logs beneath the ceiling and sitting 0.567 beneath it across the size cut. Its boundedness on these rows is measured rather than guaranteed, and every claim below that needs the guarantee is made about the forest.

Flexibility costs the tail, not the centre. A polynomial ladder in the log features (degrees 1–3) is flexible but still extrapolates. Degree 2 is *not* meaningfully worse than degree 1 on a typical machine: median 0.238 against 0.216, better on 8 of 13, paired interval $[-0.013, +0.237]$ containing zero. What flexibility costs is the tail. Degree 1's worst machine is 0.289; degree 2's is 1.083 and degree 3's is 4.601, both on C-Mod. A bounded range is simultaneously why neither ensemble extrapolates and why neither blows up: their worst cases, 0.686 and 0.857, sit far below degree 3's.

Read together, these three diagnostics invite a summary this paper does not endorse. Every flexible model scored so far is also either bounded, like the two ensembles, or divergent, like the cubic, so nothing above can tell apart "flexible models fail out of distribution" from "models that stop trending outside their data fail out of distribution". The two make identical predictions about everything in Table 1. Section 13 builds the model that separates them and finds for the second, so the reader should carry the flexibility reading as provisional from here rather than as the paper's conclusion.

5.2 The inversion under every choice Table 1 leaves open

Table 1 attributes a difference to the split, so anything else that differs between its two columns is a confound. Three things do. Grouped CV scores all 6228 rows while leave-one-out scores only the 13 machines large enough to hold out. Grouped CV pools out-of-fold predictions into one log-RMSE over every row, where JET and AUG dominate, while leave-one-out averages per-machine scores, where every machine counts once. And JET and JET-ILW are one physical tokamak either side of a wall change, as are AUG and AUGW, so holding out JET-ILW while training on JET does not hold out an unseen device.

Table 3 varies all three. The forest leads the power law under every cross-validation cell and trails it

Table 4: Three questions of increasing difficulty, scored in log-RMSE. The *larger machine* column is the ITER-size-matched cut. Reading down it rather than across the table is the point: the two models that win the first column lose the third to a constant predictor.

Model	shot	machine	larger machine
IPB98(y,2)	0.199	0.188	**0.194**
ridge, log-linear	0.181	0.214	**0.278**
random forest	0.128	0.465	**0.938**
hist grad. boosting	0.130	0.359	**1.072**
mean baseline	0.869	0.994	1.459

under every leave-one-out cell, so the inversion is a property of the split rather than of the population or the aggregation. Collapsing the variants makes the forest's failure worse rather than better: over 11 physical devices its mean gap over the power law grows from +0.251 to +0.315, with a paired 95% interval of $[+0.221, +0.408]$ against $[+0.157, +0.342]$ over the labels, and it loses on 11 of 11. The published law's score is almost unmoved by any of it, 0.188 to 0.184 per unit.

A newer published law. IPB98(y,2) is the ITER reference but not the most recent scaling fitted to this database family. ITPA20 and ITPA20-IL [2] were derived from DB5.2.3 and weaken the size dependence considerably, so they are the obvious second reference. Evaluated analytically on the rows analysed here, ITPA20 scores 0.265 over all rows, 0.267 per machine and **0.272** across the ITER-size-matched cut; ITPA20-IL scores 0.393, 0.446 and 0.363. Both are worse than IPB98(y,2) on this cleaning, and two caveats belong with that: each was fitted to a different selection out of DB5.2.3, and each specifies an elongation this database does not supply under that definition, so their absolute level here is not a fair test of either law. What survives both caveats is the comparison this paper is about. At the ITER-size-matched cut ITPA20 lands at 0.272 against the random forest's 0.938 and the gradient booster's 1.072, factors of 3.4 and 3.9, and essentially level with the blind refitted power law's 0.278. Despite the definition mismatches, both published power-law forms score substantially below the saturating tree models at that cut. Like IPB98(y,2), neither is blind.

Tuning the flexible models. Every score above uses library defaults, which invites the reading that the ensembles lost because nobody tried to make them win. We removed that reading by giving both a grid search nested inside each training fold: three grouped inner folds by discharge over the training rows only, so the held-out discharge fold, machine or size cut never takes part in choosing a hyperparameter. The forest searched `max_features` over $\{1.0, 0.5, \mathrm{sqrt}\}$ and `min_samples_leaf` over $\{1, 5\}$, the booster `learning_rate` over $\{0.05, 0.1\}$ and `max_leaf_nodes` over $\{15, 31\}$.

The search does real work on the forest and changes nothing about the conclusion. It moved off the default in 13 of the 19 folds, and it helps where the paper says a flexible model should help: cross-validation improves from 0.128 to 0.126 and the leave-one-machine-out mean from 0.465 to 0.403. At the ITER-size-matched cut it makes the forest *worse*, 0.938 to 1.121, against the power law's 0.278. The booster's search returned scikit-learn's defaults in all 19 folds, so its numbers are unchanged. Tuning therefore widens the interpolation margin that later inverts and does not repair the extrapolation failure, which is what the training-target bound predicts: no setting of these hyperparameters moves a ceiling that comes from averaging training targets.

The win counts support an exact test. Under a null that assigns each unit independently to either model with equal probability, losing 13 of 13 has two-sided $p = 2.4 \times 10^{-4}$ and losing 11 of 11 has $p = 9.8 \times 10^{-4}$. The device-collapsed count is the one to quote, since the 13 database labels are not 13 independent devices, and the same non-independence is a reason to prefer these counts to the 13-machine bootstrap.

6 A size-extrapolation stress test matched to ITER

Holding out JET still leaves twelve machines spanning much of its range, so leave-one-out extrapolates in identity while interpolating in size. ITER's major radius is 6.2 m against 3.40 m for the largest row here, a factor of 1.82. That factor is available inside the database: training on the 14 smallest machines (up to DIII-D, 1.865 m, 3498 rows) and predicting the 4 largest (up to JT-60U, 3.40 m, 2730 rows) demands a ratio of 1.823 against ITER's 1.824. The cut is selected by proximity to that ratio rather than by eye, so it is a property of the data.

At the cut the power law scores 0.278 against the analytic law's 0.194 and a constant predictor's 1.459. The random forest scores 0.938 and the gradient booster 1.072, so both sit nearer the constant than the power law in absolute error. Asked the question a scaling law exists to answer, the best cross-validated models land closer to a constant than to the published law they beat by 36% under cross-validation. The ordering is identical on all three individually scored held-out machines, so the pooled number is not an artifact of JET's weight. Four machines sit above the cut but only three are scored individually: TFTR contributes 2 rows, far below the 30 required to report a machine on its own, and enters only the pooled number. Dropping the spherical tokamaks moves the random forest from 0.938 to 0.936, so the effect is not driven by including the spherical devices; other shape or regime covariates that track size are not ruled out by this control.

7 A power law with a bounded correction

The diagnosis implies a cure. Fit the power law, learn a correction on its log residuals, and damp it by λ:

$$\log \tau = \underbrace{f_{\text{ridge}}(x)}_{\text{unbounded, log-linear}} + \lambda \underbrace{g(x)}_{\text{correction on residuals}} . \tag{2}$$

At $\lambda = 0$ this is exactly the ridge row of Table 1. Anchoring a learned correction on a frozen power law is also the design used in concurrent work on this database [16], which reports improved stability under parameter-defined distribution shift. This section makes no claim on that idea. What is varied here is the one property the diagnosis of Sec. 5.1 points at, the long-range behaviour of the correction, and the surrounding contribution is the machine-level validation reversal, its mechanism, the size-matched stress test, the constrained-exponent comparison and the interval collapse, none of which is a residual-learning result. The two correction families are chosen to contrast the long-range behaviour the diagnosis implicates. They differ in other ways as well, and the controlled version of this comparison is Sec. 13.1. One is a degree-2 log polynomial, which is unbounded; the other is depth-2 boosted trees, which as measured below never leave the residual range they were trained on. The second is an *empirically* bounded corrector rather than a provably bounded one, for the reason given in Sec. 5.1. Clipping g to the range of training residuals closes that gap at no cost. Built standalone with and without the clip, the two score identically under all three splits, 0.150, 0.249 and 0.189, and the clip binds on none of the held-out rows across the size cut. Those absolute levels are not the swept hybrid's, since the standalone build fixes $\lambda = 1$ and rebuilds the pipeline rather than reusing the sweep; the claim is the paired comparison, and it is that the empirical bound and the provable one pick out the same model here. The mechanism below can be read either way.

Across the ITER-size-matched cut the two go opposite ways. Both start at 0.278; the polynomial correction climbs to 0.356 and the bounded correction falls monotonically to **0.206**, better than plain ridge on all three held-out machines and by the widest margin on JT-60U, the most distant (0.440 to 0.281). Among the models scored at that cut it has the lowest held-out error of anything fitted without the held-out rows, 26% below plain ridge and 4.6× below the random forest, and within 6% of IPB98, which was fitted with those machines included. The constrained fit of Sec. 9, introduced later, is lower still.

The mechanism is measurable. Fitted on the 14 small machines, the base power law is *biased* on the held-out ones: mean log residual -0.218, so it over-predicts by about 20%, while its scatter (0.172) is no worse than in-sample (0.187). The tree correction never leaves the range it was trained on, on any held-out row, and inside that bound it points the right way and is largest where the bias is largest. The same bounded range that makes these ensembles useless as predictors of τ on a larger machine limits how much extrapolation error the correction can contribute, because the quantity they are bounded on is no longer a target that grows with machine size but a residual centred on zero. Bounded is not the same as right: a correction confined to that range can still point the wrong way, and what is measured here is that on these rows it does not.

Two limits should be stated. Cross-validation selects $\lambda = 1$ in both families, while the rung best on a held-out machine is $\lambda = 0$; nothing in the CV signal points at it. And scored at every rung of the size sweep rather than only the matched one, the bounded hybrid wins at 5 of 8 well-powered cuts, not all, with no rule over those eight points separating wins from losses. The claim that survives is narrow: at the cut matching the jump to ITER, and robustly across all nine correction settings tried there, a bounded correction beats the power law and the reason is understood.

Table 5: Empirical coverage of nominal 90% split-conformal intervals, and the multiplicative half-width under leave-one-out.

Model	CV	LOMO	ITER cut	width
IPB98(y,2)	90%	89%	88%	1.40×
ridge, log-linear	90%	83%	70%	1.33×
bounded hybrid	90%	64%	76%	1.26×
hist grad. boosting	91%	45%	**0%**	1.23×
random forest	91%	35%	**3%**	1.23×

8 Calibrated intervals and their collapse

For a next-step device the point error is not the deliverable; the interval is. We calibrate split-conformal intervals [3] on log residuals, holding back 25% of *discharges* to calibrate. Holding out whole discharges keeps a calibration row and a test row from coming out of the same shot, but the conformity unit is still the row, and rows arrive in correlated clusters of quasi-stationary slices from one discharge. The standard finite-sample guarantee assumes exchangeable conformity scores and clustered rows do not supply that, so every coverage number below is empirical row-level coverage rather than a guaranteed rate. A discharge-level conformity score, or a cluster-aware conformal method, would be the way to recover a guarantee, and neither is used here.

That proviso is the result. Empirical coverage is near nominal under grouped CV, where every model lands within a point of 90%, and that control is what licenses reading the rest of Table 5. It is a useful in-distribution control rather than a verified guarantee: clustered row-level conformity scores do not satisfy the finite-sample exchangeability condition either, so what the CV column shows is that the construction behaves as intended when the test rows come from machines the model has seen. It fails by construction when the test rows come from a machine the model never saw; that exchangeability is the binding assumption, and that repairing it requires knowing the shift, is the subject of the weighted-conformal literature [18], and calibrated uncertainty degrading under dataset shift is a general finding rather than a fusion one [19]. The random forest's 90% interval then covers 35% of rows on an unseen machine and 3% across the ITER-size-matched cut; the gradient booster covers none of the 2730 rows at all.

The widths barely move: no model's interval changes width by more than 1.5% between the two arms, because the half-width is set by calibration rows drawn the same way in both. The intervals do not become vague out of distribution. They stay the same size and miss.

Coverage collapses along the same axis the errors grow, but only for the trees: $\rho = -0.77$ for the random forest and -0.54 for the gradient booster against $+0.27$ for the power law, mirroring the per-machine error correlations of Sec. 5.1 with the sign flipped. The power law's intervals are somewhat too narrow everywhere rather than catastrophically too narrow far away, and for a next-step device that distinction is most of what matters, because distance is the one thing known in advance. None of this is a defect in conformal prediction. It is a measurement of an assumption being false.

9 One line of dimensional analysis

Every model to this point learns its functional form from the data. None is told any physics. That is worth revisiting, because the field has known since Connor and Taylor [5] that a confinement scaling law is not free: requiring it to be expressible in the dimensionless plasma variables imposes *linear equality constraints on the exponents.* A physics assumption is therefore a matrix C, and the fit becomes

$$\min_b \|Xb - y\|^2 \quad \text{subject to} \quad Cb = d, \tag{3}$$

which the KKT system solves in closed form. Nothing new is needed to run the experiment: the solver is the one written by hand for Sec. 4.

We derive the constraints in code from the definitions of ρ^*, β and ν^* rather than transcribing exponents from a paper, which makes the derivation checkable against something external. Written as exponents on (length, field, temperature, density),

$$\rho^* \sim T^{1/2} B^{-1} L^{-1}, \qquad \beta \sim n\,T\,B^{-2}, \qquad \nu^* \sim n\,L\,T^{-2}, \tag{4}$$

and a law expressible in whichever of these a model holds fixed must be invariant under every scale transformation that leaves those groups unchanged. Each such transformation contributes one linear

Table 6: Fitting under the Connor–Taylor constraint hierarchy. The only difference between the first and last rows is the constraint: same features, same solver, no hyperparameter.

Model	in sample	CV	LOMO	ITER cut
power law, collisionless	0.1818	0.182	0.206	**0.183**
IPB98(y,2), analytic*	—	0.199	0.188	0.194
bounded hybrid (Sec. 7)	—	0.151	0.246	0.206
power law, electrostatic	0.2245	0.225	0.241	0.237
power law, Kadomtsev	0.1808	0.181	0.210	0.254
power law, unconstrained	0.1808	0.181	0.214	0.279

*fitted on this database including the held-out machines; not a blind model.

equation in the exponents, so the admissible set is a surface and the number of constraints is set by how many groups are fixed:

constraint	groups held fixed	rows of C
Kadomtsev [11]	ρ^*, β, ν^*	1
collisionless	ρ^*, β	2
electrostatic	ρ^*	3

Fixing fewer groups admits more transformations and therefore imposes more constraints, which is why the hierarchy runs the way it does. C and d are built by `dimensional.constraint_matrix` from Eq. (4) alone, and their numerical rows are in `results/dimensional.json`. IPB98(y,2) was published in 1999 and was never told about these surfaces. It lands on the Kadomtsev surface [11] at a distance of 0.00096 and on the collisionless surface at 0.0045, both inside the rounding of its own two-decimal exponents. A transcription error would not reproduce that.

At the ITER-size-matched cut the collisionless power law scores **0.183**, the lowest held-out error of any model fitted here without using the held-out rows (Table 6). "Blind" throughout describes the fit and not the search: no coefficient ever saw a held-out row, but the constraint rung was chosen after seeing how each scored at this cut, so the cut has taken part in model selection even though it never entered a fit. It beats the bounded hybrid of Sec. 7, which was built specifically for that cut, and it beats the analytic law that was fitted with those machines included. It has no hyperparameter and nothing to tune.

The two constraints behave differently. The Kadomtsev constraint is *free*: at 0.1808 in sample it is identical to the unconstrained fit to four decimals, because the data already satisfies it unaided. It nonetheless beats the unconstrained fit at 15 of 15 cuts in the size sweep, since a constraint the data satisfies on average still stops the fit wandering when the training half is small. The collisionless constraint costs 0.001 in sample and wins at 8 of 8 well-powered cuts. Push one rung further to a beta-independent law and it degrades sharply, to 0.237. There is an optimum in the middle of the hierarchy, and nothing in this analysis predicted where it would fall; as with the hybrid, cross-validation does not select it. Every constrained fit scored slightly worse than the unconstrained one under the cross-validation used here, so CV supplies no signal pointing at the constraint that transfers best. That is an observation about these folds rather than a theorem: a hard equality constraint can only match or worsen the *training* residual, and nothing forces it to worsen a held-out score.

It matters that this is a constraint and not a prior. Supplying the same physics as a Gaussian prior instead, shrunk along the weak singular direction of Sec. 4, is worth much less. Targeting the shrinkage beats isotropic shrinkage at every matched penalty strength, which is a real effect, but nothing in that family beats simply adopting IPB98's published exponents outright. The hard constraint used here restricts the fit to a surface the Connor–Taylor similarity assumptions permit, where the Gaussian prior tested here favours a location and a direction in parameter space, and in this experiment the surface transfers better. That is a statement about these two families rather than about priors in general: a prior can favour a subspace, and approaches a hard constraint as its variance goes to zero.

10 Repairing the intervals, and the limit of the repair

The diagnosis in Sec. 8 makes a falsifiable prediction. If the interval collapse is really about exchangeability rather than about the models, then calibrating on held-out *machines* instead of held-out discharges should repair it, and should repair it only as far as that unit reaches. Both halves are tested below.

Table 7: Empirical coverage of nominal 90% intervals under three calibration schemes, on a held-out machine and across the ITER-size-matched cut.

	split conformal		by machine		+ distance	
Model	LOMO	cut	LOMO	cut	LOMO	cut
random forest	35%	3%	**88%**	29%	88%	40%
hist grad. boosting	45%	0%	**90%**	4%	89%	13%
ridge, log-linear	83%	70%	91%	77%	92%	82%
power law, collisionless	85%	91%	91%	**95%**	92%	92%
IPB98(y,2)*	89%	88%	89%	89%	89%	87%

The machine-level scheme uses an inner leave-one-machine-out loop, and nothing in it touches the target machine. For each target m: remove m; the remaining machines are the training set. Inside that training set, hold out each machine k with at least 30 rows in turn, fit on the others, and record $|\log\hat{\tau} - \log\tau|$ on k together with each of its rows' Mahalanobis distance from that inner fit set. Pool those residuals into the conformal quantile. Then fit the final model on all the training machines and score m once. Every conformity score therefore comes from a machine its own model had not seen, which is the unit the test rows differ by, and no row from m enters any fit or any quantile. The distance-scaled variant fits $\sigma(d) = \exp(c_0 + c_1 d)$ to those calibration residuals by least squares on $\log|r|$ and divides the scores by it.

Both arms behave as the diagnosis predicts. On a held-out machine, empirical row-level coverage returns close to nominal: every model lands within two points of it, tree ensembles included, and the random forest goes from 35% to 88%. Across the size cut none does. Neither recalibration procedure tested restores near-nominal coverage there, because every calibration machine is smaller than every test machine, and scaling the interval by extrapolation distance recovers only part of the gap (3% to 40% for the forest). A repair that stops exactly where the diagnosis says it should is stronger evidence for the diagnosis than one that worked everywhere would be.

Among the fitted blind models, the constrained power law of Sec. 9 is the only one in Table 7 whose plain split-conformal coverage stays near nominal at the ITER-size-matched cut, and its intervals hold there under every scheme. IPB98(y,2) also holds at 88%, but it was fitted with those machines included.

11 Robustness on rows the standard analysis set excludes

Everything above rests on one file, which is the honest ceiling on all of it. The same OSF project publishes the full database revision, DB5.2.3, with 14153 rows against STD5's 6228, and we pin it by its own SHA-256. Matching on (tokamak, shot, time) shows STD5 is an ELMy-H-mode quality selection out of it and leaves two populations this study had not analysed: **5358 H-mode rows STD5 does not contain** (12 machines, zero row overlap) and **3860 ohmic and L-mode rows** in a different confinement regime, which we score against ITER89-P [6] because an H-mode law is the wrong baseline for an L-mode plasma.

This replication has one way to be silently fake. If the row match against STD5 fails, every STD5 row stays in the "disjoint" arm, the arm becomes a superset of the data Sec. 5 was computed on, and it reproduces Sec. 5 by construction. We therefore check the matcher positively as well as negatively: STD5 matched against itself must overlap completely, and the disjoint arm must share zero rows.

The ranking inverts in both arms. On the disjoint H-mode rows the best cross-validated model beats the published law by 42%, a margin of the same size as the 36% over that law reported above and computed on rows it never saw, and the best model on an unseen machine is IPB98(y,2). On the ohmic and L-mode rows the cross-validation gain over ITER89-P is 67% and the best model on an unseen machine is the collisionless power law of Sec. 9. Counting machine–model pairs where a tree ensemble loses to the published law on an unseen machine: 19 of 24 and 10 of 10.

Row disjointness is not discharge disjointness, and the paper's own argument says the discharge is the unit that matters. Of the 5358 H-mode rows, 425 (7.9%) are slices of one of 336 discharges that STD5 samples at a different time. Dropping those discharges outright leaves 4933 rows across 2864 discharges that share no shot with STD5 at all, and the arm survives: the best cross-validated model still beats the published law, by 39% against 41% on the full arm, and the best model on an unseen machine is still IPB98(y,2), with the tree ensembles at 0.526 and 0.552 against the power law's 0.332. The reversal does not depend on the shared discharges.

So the reversal is not an artifact of the standard set's selection criteria, and not a property of ELMy

H-mode or of IPB98(y,2) specifically. It is emphatically *not* an independent database: both arms come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone.

12 A locked prediction for three machines that have no data

Everything above is retrospective. To make the disagreement falsifiable rather than merely described, we record what each model predicts for three real devices, with intervals, a date, and a digest over the input rows so that a later edit leaves a mark. The parameter sets are published design values, and each reproduces the confinement time its own source quotes: SPARC to 0.6% and ITER to 2.9%.

Table 8: Predicted energy confinement time, in seconds, with nominal 90% intervals. Locked 31 August 2026 against a fixed dataset digest. Operating points are published design values: SPARC from [10] (V2 primary reference discharge), ITER from [1] ($Q = 10$ inductive baseline), and JT-60SA from its Research Plan [22] (full-current inductive scenario). The complete input vector for each device is recorded in `results/forecast.json` under the content digest quoted below.

	SPARC 1.85 m	JT-60SA 2.96 m (operating)	ITER 6.2 m
IPB98(y,2)*	0.765 [0.58, 1.00]	0.479 [0.35, 0.66]	**3.591 [2.66, 4.85]**
power law, collisionless	0.724 [0.55, 0.96]	0.428 [0.31, 0.59]	2.837 [2.09, 3.84]
ridge, log-linear	0.720 [0.53, 0.98]	0.444 [0.32, 0.62]	2.858 [2.07, 3.95]
random forest	0.136 [0.03, 0.53]	0.449 [0.23, 0.87]	**0.435 [0.19, 0.98]**
hist grad. boosting	0.305 [0.14, 0.69]	0.418 [0.24, 0.74]	0.444 [0.24, 0.84]

Table 8 is the argument of this paper in one place. On JT-60SA, which sits inside the database's size range and is already operating, all five models agree to within 15%. On ITER, 1.82× beyond it, they disagree by a factor of **8.3**.

For the random forest that gap cannot be closed by tuning the standard estimator, and Sec. 5 says why: it predicts by averaging training targets, so its output cannot exceed the largest one, which here is 1.321 s, and no hyperparameter moves that ceiling. The gradient booster is not bound by that arithmetic and lands in the same place anyway. The random forest that beats the published law by 36% under cross-validation is arithmetically incapable of returning ITER's predicted confinement time whatever it is asked, and its nominal 90% interval at ITER runs from 0.19 s to 0.98 s, containing neither the physics-informed reference predictions nor anything near them.

The JT-60SA column becomes checkable first, but not as of September 2026: the first campaign ran L-mode divertor plasmas, and its published power balance reports the energy confinement time following L-mode scaling [23], whereas the row above is an H-mode prediction scored against IPB98(y,2). Checking it needs H-mode operation at a comparable design point, which the research plan [22] places in a later campaign; we deliberately do not put a date on that here, because the schedule is not ours to assert and the prediction is locked against a parameter vector rather than against a calendar. Each row of Table 8 carries its own interval and each is falsified separately, so the range quoted here names one model rather than the set. A stationary JT-60SA H-mode near this design point reporting a thermal confinement time outside 0.349 s to 0.657 s would falsify the IPB98(y,2) reference row. The blind rows are falsified at their own bounds, which are narrower for the constrained power law (0.313 s to 0.587 s) and wider for the random forest (0.232 s to 0.867 s). Because all five intervals overlap across roughly this range, a measurement inside it separates none of them: this row tests whether any model is right, not which, and that is why the ITER column carries the argument.

Nothing above anticipated one feature of this table: SPARC sits further from the training data than ITER does, at a Mahalanobis distance of 6.9 against 4.7, despite being the smaller machine. Its 12.2 T field is far outside a database that tops out near 4 T. Size is not the only direction in which a next-step device leaves the data behind, and an extrapolation audit organised around size alone would have missed it.

Two replications outside fusion are reported in Appendix A and Appendix B: the same audit run on Kleiber's law for mammalian metabolic rate, and a predictor-count sweep on woody plants that isolates what the ranking inversion needs and the extrapolation failure does not. They are what rules out reading any of the above as a fact about tokamaks.

model	CV	held-out	ITER cut	ρ
GP, linear + RBF	**0.112**	0.218	**0.191**	-0.01
GP, RBF only	0.142	0.541	1.948	$+0.65$
GP, linear only	0.181	0.214	0.278	-0.06
random forest	0.128	0.465	0.938	$+0.85$
IPB98(y,2), not blind	0.199	0.188	0.194	-0.49
mean baseline	0.869	0.994	1.459	$+0.56$

Table 9: log-RMSE under the three splits, and the rank correlation of per-machine error against extrapolation distance. The intended difference across the first three rows is whether the kernel admits a trend that continues at long range; adding the linear term also enlarges the function class. The three span a factor of ten at the ITER-size-matched cut. IPB98(y,2) was fitted with these machines included and is a reference point rather than a blind baseline.

13 Flexibility, or long-range saturation?

Section 5.1 left two readings open, and the polynomial ladder there supports the flexibility one only for a single kind of flexibility: polynomial degree under an isotropic penalty, swept over nine decades without rescue. Every model scored to this point confounds two properties. Flexibility is how much structure a model can learn beyond a power law. *Boundedness* is what it does far outside the training data. We call a model *saturating* when its predictions stop tracking the features once the input leaves the training support, either because they cannot leave a fixed range or because they revert to a fixed value. The random forest is the strict case, provably confined to $[\min y_{\text{train}}, \max y_{\text{train}}]$; the gradient booster carries no such theorem but never left that range on any split scored here (Sec. 5.1); an RBF process reverts to its prior mean, a third mechanism with the same consequence. What the three share is the absence of a trend that continues outside the data, and that is what this section tests. The tree ensembles here are flexible and saturating; ridge is neither; a degree-3 polynomial is flexible and non-saturating but divergent, which is why its worst machine is 4.601. No model to this point was flexible, non-saturating and well behaved at long range at once.

A Gaussian process [9] supplies one, and the property is a choice of kernel rather than a tuning parameter. We fit three over the same log features, with the same optimizer on the same rows. The intended difference between them is whether the kernel admits a trend that continues at long range. Adding a dot-product term also enlarges the function class, so this is not a clean single-factor experiment and we do not read it as one. An RBF kernel decays to zero with distance, so the posterior returns to its prior mean, which centring places at the training mean: bounded. A dot-product kernel is Bayesian linear regression in the log features, which is a power law: unbounded and inflexible. Their sum is both. Hyperparameters are fitted by maximising the log marginal likelihood on a seeded subsample of each training fold, never on held-out rows; hand-chosen values are not adequate here, since a length scale an order of magnitude from the one the data supports makes the third rung look like a failure.

The control lands: the linear-kernel process and ridge agree to 0.0000 under cross-validation and on a held-out machine, and to 0.0007 at the ITER-size-matched cut, which is what licenses reading the other two rows as consequences of the kernel. The bounded rung then fails exactly as a tree does. It beats the power law under cross-validation and scores 1.948 at the ITER-size-matched cut, worse than predicting a constant, with error tracking distance at $\rho = +0.65$. The mechanism is the one in Sec. 5: a random forest cannot exceed the largest training target, and an RBF process reverts to its prior mean. Both are structural long-range behaviours rather than tuning shortfalls, and on the held-out rows this one's predictions carry half the spread of the truth.

Adding a dot-product term to that same kernel keeps the nonlinear RBF component it already had and supplies an unbounded linear trend alongside it. Flexibility is retained rather than held exactly fixed, since a linear kernel enlarges the function class as well as changing the asymptote; what the comparison isolates is the availability of a continuing trend. It gives the best cross-validated score in this work, 0.112 against the random forest's 0.128; it scores 0.191 at the ITER-size-matched cut, better than the analytic law fitted with those machines included and second only to the constrained fit of Sec. 9 at 0.183; and its error is flat against distance at $\rho = -0.01$.

13.1 The same experiment with the nonlinear component specified identically

Adding a dot-product term enlarges the function class, so the ladder above suggests the mechanism without isolating it. One rearrangement removes that objection. Write the model as a mean function plus a residual

$m(x)$ in Eq. (5)	CV	held-out	ITER cut
constant	0.142	0.541	1.948
fitted power law	**0.115**	**0.212**	**0.278**

Table 10: One RBF covariance family and fitting procedure, two mean functions. The constant-mean row reproduces the RBF rung of Table 9 to six decimals, which is the control: the decomposition is the same model rewritten. Changing only the mean-function specification moves the ITER-size-matched cut by a factor of 7.0.

process,

$$\log \tau = m(x) + g_{\mathrm{RBF}}(x), \tag{5}$$

and specify g identically in both arms: same RBF covariance family, same tuning subsample, same seed, same optimizer. Only m changes, from a constant to a fitted log-linear power law. The two fitted processes are not the same function, because each is conditioned on its own mean's residuals and each re-optimises its hyperparameters on those. What is held fixed is the nonlinear capacity available to the model and the procedure that fits it, so no part of the difference below can come from one arm having more of either.

The control lands first: with a constant mean, Eq. (5) returns 0.141533, 0.540967 and 1.947535, which is the RBF row of Table 9 to every digit computed. Nothing has changed except how the model is written. Giving the same process a power-law mean then moves the ITER-size-matched cut from 1.948 to 0.278, a factor of 7.0, and improves the cross-validated score as well, 0.115 against 0.142. Because the residual GP has the same covariance family, the same capacity and the same fitting procedure in both rows, the difference cannot be attributed to adding nonlinear model capacity.

The second row's cut score is worth reading closely: 0.277823 is plain ridge's score at that cut, to six decimals. Far outside the data the RBF residual has reverted to zero, which is what an RBF process does, and the model falls back to its power-law mean. That is the whole mechanism in one number. A saturating model reverts to something; the failure in Table 9 is that a constant mean makes that something a constant, and the repair is not to make the model less flexible but to give it something worth reverting to.

Across the model families tested here, this resolves the choice Sec. 5.1 left open: extrapolation failure tracks long-range saturation rather than flexibility alone. The set of models in Sec. 5 is measured correctly, and every flexible member of it was either bounded or divergent, so nothing in it could have separated the two properties; that is why the reading there was flagged as provisional rather than asserted. "Do not use a flexible model out of distribution" is the wrong lesson. "Do not use a model whose predictions stop tracking the features once the input leaves the training data" is the right one, and the random forest's training-target ceiling is its strictest case. This does not displace the constrained power law, which still wins the ITER-size-matched cut with nothing to tune, and the process does not beat the power law on a held-out machine (0.218 against 0.214).

A Gaussian process also supplies an interval by construction rather than by calibration, so Sec. 8's question can be put to a model that was never calibrated. At the ITER-size-matched cut, at nominal 90%, the flexible unbounded rung covers 92.5%, where split-conformal intervals give the random forest 3% and the gradient booster none. It is the only model here whose out-of-distribution intervals reach nominal without being recalibrated to do so. The bounded rung is the instructive failure: it is the one model that does become vague out of distribution, with a half-width four times any other, and it still covers 16.7%, having reverted to a prior mean nowhere near the answer. Width was never the problem.

14 Discussion

The practical consequence of Table 1 is narrow and specific. It is not that machine learning has nothing to offer confinement scaling, and not that the models scored here are badly built. It is that the comparison the field runs by default answers a question nobody is asking. Cross-validation grouped by discharge, or by shot, or by time slice, holds out rows from machines that remain in the training set. It measures interpolation within known devices. A scaling law is used to size a device that does not exist, which is extrapolation to an unknown one, and on this database the two comparisons do not merely differ in magnitude: they order the candidate models in opposite directions. A gain reported under the first is not evidence about the second.

That suggests a reporting convention rather than a modelling one. Any confinement model offered as an alternative to a power law can be scored under leave-one-tokamak-out on the same rows and the same features, at a cost of one refit per machine, and the two numbers can be printed together. Where they agree, the interpolation number is informative. Where they disagree, as here by a factor of 3.6, the disagreement is the result. Two diagnostics that need no held-out device are worth reporting alongside: the distance of the target operating point from the training distribution, which tracked per-machine error at $\rho = +0.85$ for the forest and not at all for the power law, and whether the model's prediction can leave the range of its training targets at all, which for a random forest it provably cannot.

The positive findings point the same way, and both are about the far field rather than about capacity. Constraining the exponents to a Connor–Taylor surface and adding a linear component to a Gaussian process kernel are unrelated operations, and neither makes its model more flexible; each fixes what the model does once the input leaves the data. That the two arrive at the same place is the strongest support here for reading the failure as long-range saturation rather than flexibility, and the mean-function ablation of Sec. 13.1 is the cleanest single piece of it, since it holds the function class fixed and varies only the asymptote. It should still be read as a statement about the families tested, not about flexible models in general.

For practice, the cheaper option is the stronger one out of distribution. A power law refitted on the same rows is a stronger baseline out of distribution than any tree ensemble scored here, and it costs a single least-squares solve; the estimator hardly matters, since ordinary least squares on the independent columns reproduces the reported ridge to within 0.0012 on every split. Where a flexible model is wanted, the evidence here favours one that keeps trending, and a Gaussian process with a linear mean was both the best cross-validated model in this study and the only one whose out-of-distribution intervals reached nominal without recalibration. Where a physics constraint is available, imposing it costs nothing in sample and was the best-scoring option at the size cut.

None of this is settled by one database. The reversal needs enough feature dimensionality for the flexible model to win the interpolation comparison first, which Appendix B establishes and which means a field whose scaling law has one or two predictors faces the same extrapolation hazard with no warning sign at all. The extrapolation failure itself reproduced everywhere it was tested; it is the inversion, not the failure, that requires a condition to appear.

15 Limitations

The refit population is not IPB98's, and the refit exponents should not be read as a correction to it. Two machines (JET and ASDEX Upgrade with their variants) supply 77% of rows. Only 13 of 18 machines are scored under leave-one-out; the five excluded are the smallest and most unlike the rest, so the reported gap is if anything optimistic. Thirteen machines is a small bootstrap and the paired differences are the more reliable statistic. The training-target bound of Sec. 5.1 is a theorem for the random forest and an observation for the gradient booster: boosting sums tree outputs rather than averaging targets, so nothing confines it to the training range, and what is reported here is only that it stayed inside that range on all 14 splits scored. The ITER-size-matched cut reproduces ITER's size ratio and nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient. The per-machine coverages in Sec. 8 are proportions over as few as 39 rows and carry binomial uncertainty of ten points or more; the pooled contrasts are far larger than that noise, the within-column ordering is not. The constraint hierarchy of Sec. 9 has an optimum in its middle that nothing here predicted and that cross-validation cannot select, so the result is that a constraint helps rather than that this constraint is the right one. The robustness arms of Sec. 11 are disjoint in rows but not independent in provenance: both come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone. The H-mode arm is disjoint in rows and not, as first constructed, in discharges; Sec. 11 reports the rerun with all 336 shared discharges removed, which the reversal survives. The locked forecast of Sec. 12 is a record, not a validation; only JT-60SA can be checked against measurement in the near term, that check requires an H-mode campaign rather than the completed L-mode one and we put no date on when that arrives, and it is the one device of the three that sits inside the database's size range. The replication of Appendix A has one predictor and eleven groups, and those groups are phylogenetically related rather than independent, which the comparative-biology literature handles with phylogenetically controlled regression [21] and we do not; it suffices to show the extrapolation failure is not fusion-specific and that the reversal needs a condition that dataset lacks, not to establish a rate for either. The ladder of Appendix B is one ordering of one set of predictors, so it establishes that a crossing exists and that the inversion tracks it, not that three is a threshold; its rows are a selective intersection, cutting 21084 plants to 3599 and favouring studies that measured leaves destructively, most of those plants are glasshouse-grown or from common gardens rather than wild forest, and its 14 source studies mean holding out a species sometimes holds out a laboratory. Every power-law fit here treats the engineering parameters as measured without error, which the confinement-scaling literature has long objected to because these predictors carry real measurement uncertainty. Refitting Eq. (1) by orthogonal distance regression, which spreads the error across the predictors instead, moves the exponents enormously: up to 5.6 on the inverse aspect ratio, with the major radius going from 1.58 to 5.40. It also more than doubles the in-sample error, 0.454 against 0.181. Both are symptoms of one thing rather than a better fit. With no per-variable measurement uncertainties published for this database, an errors-in-variables estimator has nothing pinning it down along the weak singular direction of Sec. 4, which carries 0.3% of the design's variance, and it runs away along exactly that direction. The honest statement is that these are least-squares exponents, that an EIV fit would differ, and that this database does not supply what computing a credible one would need. Every fit also weights rows equally, so JET and ASDEX Upgrade set most of what each model learns even though the target is transfer to a device neither resembles; refitting the power law with each training machine weighted equally does not help, moving its leave-one-machine-out score from 0.214 to 0.259, so the row imbalance is not an easy source of its advantage, though that is one weighting of one model and does not settle the question. The Gaussian-process intervals of Sec. 13 are equal-tailed posterior intervals on $\log \tau$ that include the fitted observation-noise term and are exponentiated to τ; that is the closest available analogue of a conformal interval, which is also a statement about a new observation rather than a latent mean, but the two are not the same construction and the comparison should be read as such. Those kernels are tuned by marginal likelihood on a seeded subsample of each training fold, with only the hyperparameters coming from the subsample and the posterior then conditioned on the whole fold, which is stable across the range we could afford (under 4% in every hyperparameter between 750 and 1500 rows) but is not a proof that the full-data optimum agrees; and "linear plus RBF" is one reading of "a power law with a bounded correction" among several, with an anisotropic kernel, a Matern kernel and a linear part constrained as in Sec. 9 all untested. The last of these is the obvious next experiment, since the constraint and the kernel are the two things that work here and nothing has tried them together.

16 Conclusion

On real confinement data, the model that wins cross-validation by 29% over a blind refitted log-linear power law loses to that same power law on all 13 held-out database labels and on all 11 physical devices once the JET and ASDEX Upgrade variants are collapsed, and to the published IPB98(y,2) reference as well, though that reference was fitted with those machines included and is not a blind comparator. Three separate diagnostics point at that failure and none of them is a tuning fix; the mechanism they converge on is long-range saturation. At the size extrapolation that separates this database from ITER, the best cross-validated models land closer to a constant predictor than to the power law, and their nominal 90% intervals cover 3% and 0% of the held-out rows at unchanged width. In these tests the power-law form survives leaving the data behind despite fitting the observed rows less accurately under conventional cross-validation. A bounded correction on its residuals improves it in the one direction a next-step device lies in, and the bounded range that disqualifies trees as predictors is what limits how much extrapolation error that correction can contribute.

The repair that scored best, though, is not additional model flexibility at all: it is a dimensional constraint on the existing power law. Constraining the exponents to the surface implied by the collisionless Connor–Taylor similarity assumptions gave the lowest observed error of any fit tested here at the ITER-size-matched cut, cost 0.001 in sample, and has nothing to tune. That rung was selected after the held-out scores at this cut had been seen, as Sec. 9 states, so it is the lowest error among the surfaces examined rather than a prediction that this surface is the right one. Supplying the same physics as the Gaussian prior tested here is worth substantially less: the constraint restricts the fit to a surface those assumptions permit where the prior only favours a location within parameter space, and here the surface transfers better. The inversion itself survives several separate robustness checks: rows the standard set does not contain, a second confinement regime, collapsing the 13 database labels into 11 physical devices, and either aggregation of the score. It is therefore a property of how these models are validated rather than of one file or one bookkeeping choice. What the models disagree about is recorded rather than argued: on a machine inside the database's size range they agree to within 15%, and on ITER they differ by a factor of 8.3.

Two results temper the diagnosis and neither weakens the practical advice. Varying only the number of predictors shows that the extrapolation failure persists at every predictor count tested while the ranking inversion does not: the inversion needs the flexible model to win the interpolation split first, so a field whose scaling law has one predictor faces the same danger with no warning at all. And separating flexibility from long-range saturation shows the failure belongs to the second. A bounded RBF process fails as the trees do, while adding an extrapolative linear component to that process gives the best cross-validated score here, holds up at the ITER-size-matched cut, and is the only model in this work whose out-of-distribution intervals reach nominal without recalibration. That comparison enlarges the function class as well as changing the asymptote, so the cleaner test is the ablation that fixes the nonlinear component's specification and varies only the mean function. It moves the ITER-size-matched cut by a factor of 7.0, and the better arm lands exactly on the power law's score there, because far from the data its correction has reverted to zero and left the trend behind. The lesson is not that flexible models cannot be trusted beyond their data. It is that a model whose predictions stop tracking the features once the input leaves the training data is structurally unsuited to a problem where the target itself has to leave that range. The two are easy to confuse because the flexible models that win the conventional interpolation comparison here are the saturating ones, while the non-saturating polynomial control becomes unstable in extrapolation instead.

Acknowledgments

This work uses the ITPA Global H-mode Confinement Database, assembled and maintained by the ITPA Confinement Database and Modelling Topical Group and by the teams operating the contributing tokamaks. The author thanks them, and the authors of Ref. [2], for releasing the standard analysis set publicly: an independent recomputation of this kind is possible only because they did. The replications outside fusion use the BAAD compilation [7], released under CC0, and the metabolic-rate compilation deposited on Figshare that Appendix A describes. Neither the ITPA group nor any contributing team has reviewed or endorsed this analysis, and its conclusions are the author's alone.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests

The author declares no competing interests.

Author contributions

Sole author. L.S. designed the study, wrote the analysis code, performed the analysis, and wrote the manuscript.

Use of AI tools

An AI assistant (Anthropic Claude) was used in developing the analysis code, in carrying out parts of the analysis, and in drafting and editing this manuscript. No AI system is an author. The author designed the study, verified every reported number against the generated artifacts, and takes full responsibility for the content, the conclusions and any error in them.

Data and code availability

The HDB5 STD5 dataset is third-party data obtained from OSF (`https://osf.io/drwcq`) and is not redistributed here. Every input is verified against a pinned SHA-256 on load, so each number above refers to one specific set of bytes:

Table 11: Kleiber's law under the same three splits. Scores are log-RMSE.

Model	CV, by species	leave-one-order-out	widest mass cut
Kleiber, exponent 3/4	0.396	0.440	**0.374**
power law, free exponent	0.374	0.432	0.496
hist grad. boosting	0.418	0.487	0.889
random forest	0.437	0.517	0.710

file	SHA-256
`hdb5_std5.csv`	`67601c2da5c51f90cf6298ff499cccc7` `4d09ac80c2b98c7dde0d8db3ebb9ac5b`
`hdb5_db523.csv`	`7a48e34379663e3e298924990f05cd8f` `16b8581516bcfa7bb8f438f83ae80ab6`
`allometry_bmr.txt`	`ab22d9ae8f35e96d7fbf5d8e53053d64` `7f0594d74f8917888169993ce668571e`
`baad_data.zip`	`0375f012475658c039e3dcef398951e8` `f68cdeb5430eec8cd65548965496cfe2`

The locked forecast of Sec. 12 additionally carries a content digest over its input rows, so a later edit to the operating points leaves a mark:

```
960d537ee3df8780afd2054e51d01a268
f8803aca64a830dba37fd1ff5319192
```

All code, the full writeup with every table and limitation, and the scripts that regenerate every number and figure in this paper are at `https://github.com/lsalcedo100/FusionFlux`. The analysis code is pinned as firmly as its inputs: every number and figure above was produced at commit

```
187b49aacb3f8c15f993b4e3d69634d688114219
```

Each analysis script also writes the dataset fingerprint it verified into its own JSON output under `results/`, so a regenerated number carries the provenance of the bytes it was computed from.

A The same audit on a scaling law from another science

Every result to this point rests on ITPA data, so the last question is whether any of it is about fusion. Mammalian basal metabolic rate against body mass is the same object and the older one: Kleiber [4] found in 1932 that metabolic rate scales as mass to the 3/4. Whether the true exponent is 3/4 or 2/3 has been argued for decades and we take no position; 3/4 is used here as the canonical historical reference law a fitted model has to beat. Taxonomic *order* is the unit a species belongs to, as a tokamak is the unit a discharge belongs to, and *body mass* is the axis a next instance lies beyond, as machine size is. The data is Supplement 1 of a metabolic-rate compilation deposited on Figshare (article 3549807), pinned by SHA-256 as the HDB5 files are, giving 541 species records across 11 orders whose median masses span 342x. The analysis runs through the domain-agnostic module released with this work rather than a copy of the fusion pipeline, so it is the procedure a reader would apply to their own data, and the problem has a *single predictor*: no model can win by finding a better combination of features, so the only thing separating them is functional form. Refitting the exponent freely gives 0.687 against Kleiber's 0.75, so the constraint is not free here either.

The extrapolation failure reproduces. Both tree ensembles are worse than both power laws at all 8 mass cuts, and the random forest loses to Kleiber on 9 of 11 held-out orders. Two signatures reproduce with it. Tree error rises with extrapolation distance, at $\rho = +0.64$ and $+0.65$ against $+0.39$ and $+0.45$ for the power laws, which is the same sign as Sec. 5.1's but a much narrower contrast, since here the power laws rise with distance too. And the training-range bound binds at every cut.

The ranking reversal does not, and that bounds the claim. The trees never win the easy split either: under cross-validation by species they score 0.437 and 0.418 against the free power law's 0.374. With no inflated margin there is nothing available to invert. The HDB5 gain came from a flexible model exploiting structure across nine features; with one predictor and a relationship close to a straight line in logs, a tree has far less to exploit and does not buy the interpolation win that later reverses. The reversal requires enough feature dimensionality for the flexible model to win interpolation first, which no single database could have shown.

predictors	interp. gain	held-out gain	reversal
diameter	−1.7%	−10.5%	no
+ height	−0.6%	−10.8%	no
+ leaf area	+1.8%	−13.1%	yes
+ leaf mass	+6.8%	−13.4%	yes

Table 12: The best tree ensemble's margin over the log-linear power law, on 3599 plants held fixed across the four rungs. Positive favours the trees. The interpolation column crosses zero; the extrapolation column does not, and grows slightly against them.

The constraint result reproduces in direction, not in strength. Kleiber's published exponent costs +0.023 under cross-validation and wins the widest cut, 0.374 against 0.496, where the held-out order is 69x heavier than the training median: Sec. 9's trade exactly. But it is weaker, winning at 4 of 8 cuts rather than 8 of 8, decisively at the extremes and narrowly losing in the middle. That a constraint helps out of distribution replicates; that it helps reliably does not.

B The reversal's precondition, measured directly

Appendix A leaves an asymmetry. The extrapolation failure reproduced in full on Kleiber's law; the ranking inversion did not, and the explanation offered there was a conjecture: with a single predictor a tree has little to exploit, never wins the interpolation split, and so has no advantage to lose. Two datasets cannot test that, because HDB5 and the mammalian compilation differ in feature count and also in science, group structure, sample size and noise.

A third dataset can, if the feature count is varied inside it while everything else is held fixed. The Biomass And Allometry Database [7] records, for individual plants, a nested set of measurements: basal diameter, height, leaf area and leaf mass. Taking the rows complete in all four gives 3599 plants across 53 species, 33 of them with the 30 individuals we require to hold a species out, spanning 36x in diameter across species medians and over seven orders of magnitude in mass. It is released CC0 and pinned by SHA-256 as the other deposits are. Species plays the part of tokamak, diameter the part of machine size, and the branching-network prediction that mass scales as diameter to the 8/3 [8] the part of IPB98(y,2): an exponent derived from theory rather than fitted, which refits here to 2.512 and costs 0.5641 against 0.5780 in sample when held.

Each rung of the ladder is a prefix of the next and all four are scored on the same rows, so the only quantity that changes across them is how many predictors the models may see. The split structure matters and is easy to get wrong: grouped cross-validation by discharge keeps every machine on both sides, so its analogue keeps every species on both sides. Grouping the cross-validation by species instead would be a second leave-one-group-out, and against it no reversal can appear at any dimension.

Table 12 separates two statements that had been travelling as one. The interpolation margin rises monotonically and crosses zero between two and three predictors. The extrapolation margin does not shrink as the trees improve; it widens, and the power law wins the held-out species at 4 of 4 rungs. The inversion therefore appears at 3 predictors and every rung above, which is exactly where the interpolation margin turns positive. What the ladder shows is that the inversion tracks the interpolation margin, not that three is a threshold: each new column adds real biological information as well as a dimension, and leaf mass in particular is closely related to the target, so the rungs differ in more than their count.

So the extrapolation failure holds in tokamaks, mammals and trees alike, at every predictor count tested here. The *inversion* is conditional, and its condition has nothing to do with the failure: it is only whether the flexible model had enough to work with to win the comparison a practitioner runs first. The inversion is therefore a symptom of the extrapolation failure rather than the failure itself. Below three predictors the hazard is identical and the light is off.

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Figure 1: Interpolation against extrapolation. Top: each model under both splits, with bars giving the 95% interval over the 13 held-out machines; the lines crossing is the result. Middle: per-machine error against Mahalanobis distance from the training data, rising strongly for the random forest, more weakly for the booster, and flat for the power law. The circled machine is the second failure mode, where the true confinement times run above anything the random forest can output. Bottom: the flexibility ladder, where what grows with flexibility is the tail rather than the centre; the circled points are the two tree ensembles, both observed here to stay inside the training-target range, though only the random forest is provably bounded to it.

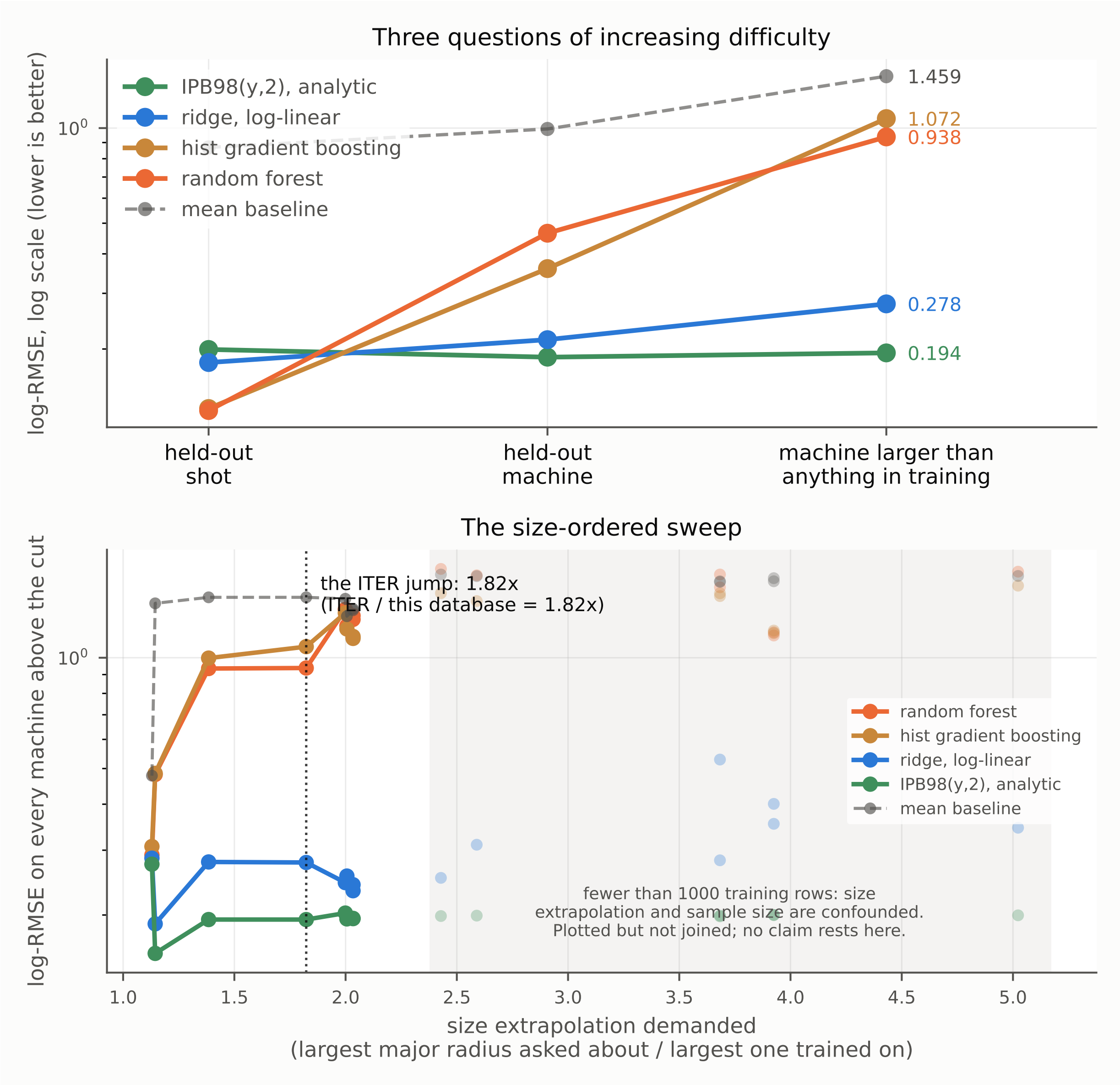

Figure 2: Size-ordered extrapolation. Top: the escalation across three splits of increasing difficulty, where the tree ensembles collapse toward the mean baseline. Bottom: the sweep across every cut, with underpowered cuts drawn faint and excluded from claims. The power law holds across the sweep and the trees do not recover.

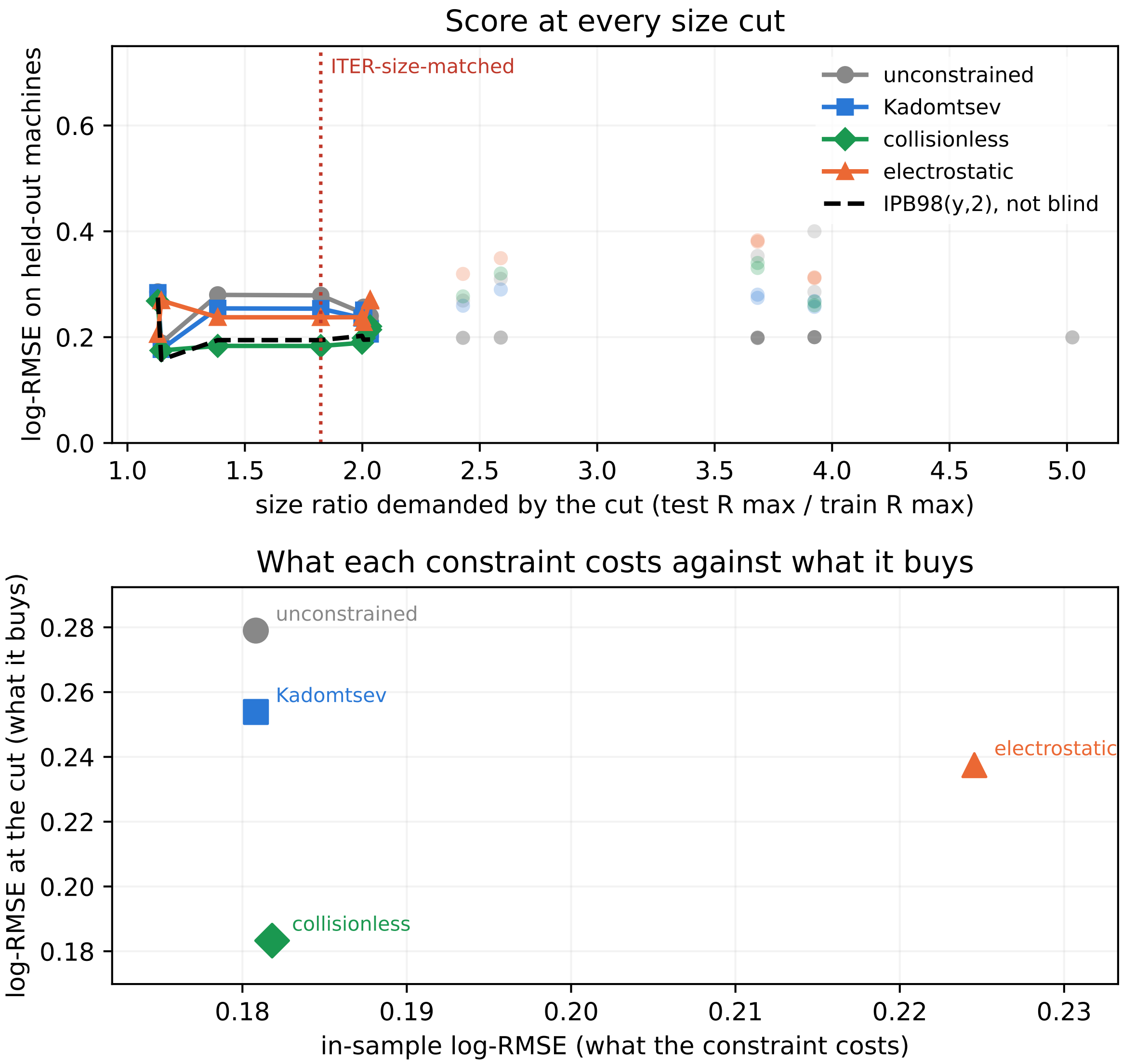

Figure 3: Cost against benefit across the size sweep. Top: the score at every size cut for the constrained and unconstrained fits, with underpowered cuts drawn faint and excluded from claims. Bottom: what each constraint costs in sample against what it buys at the ITER-size-matched cut. The constraint costs almost nothing where the data is plentiful and pays where it is not, and more physics is not monotonically better: the hierarchy has an optimum in its middle, at the collisionless rung.

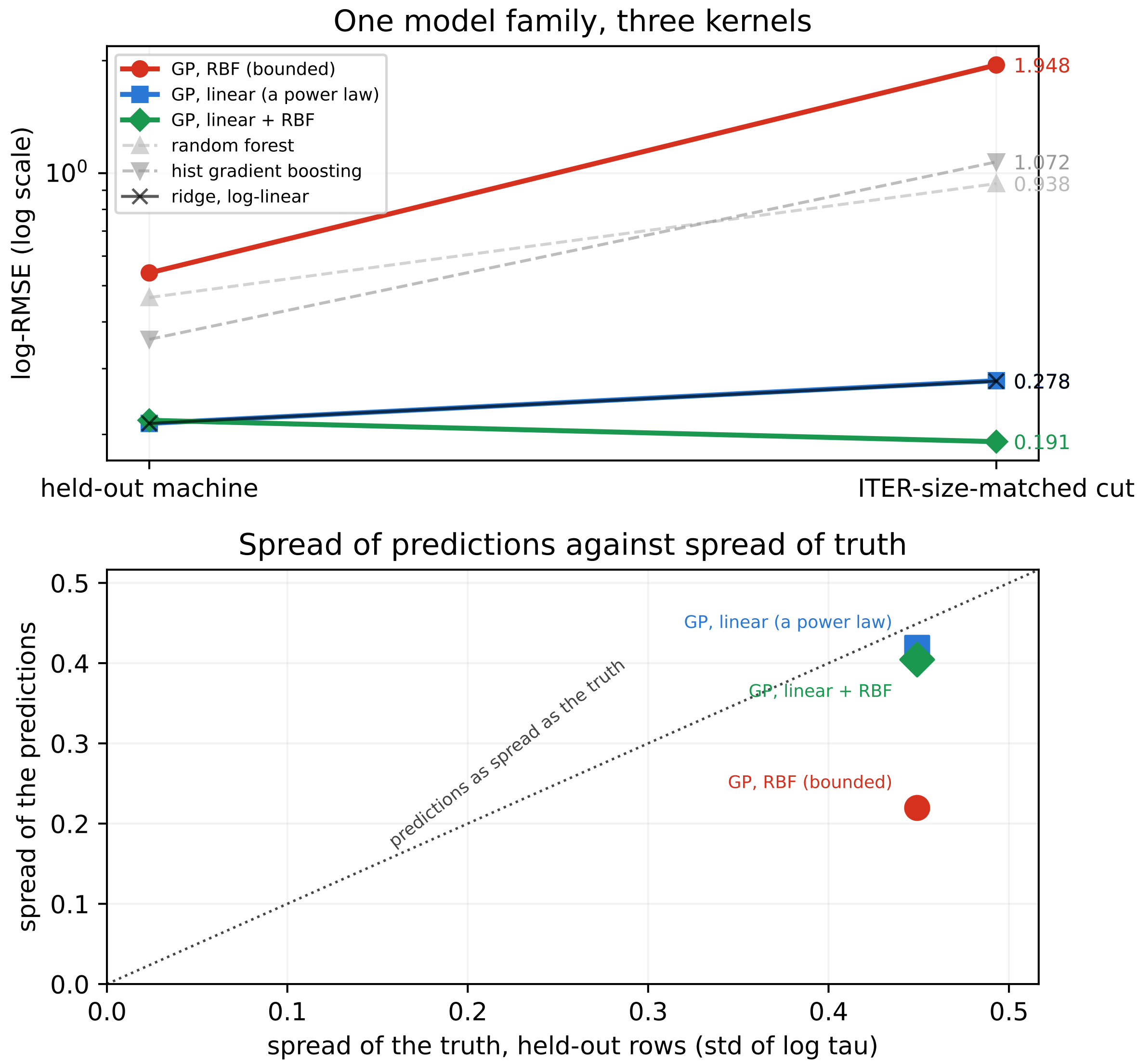

Figure 4: One model family, three kernels. Top: log-RMSE on a held-out machine and at the ITER-size-matched cut. The kernels differ in whether they permit a trend that continues at long range; adding the linear component also enlarges the function class. Bottom: spread of the predictions against spread of the truth on the held-out rows. A model that has reverted to its prior mean sits near zero however good its in-range fit was, which is the tree ceiling of Sec. 5.1 reached by a different route.

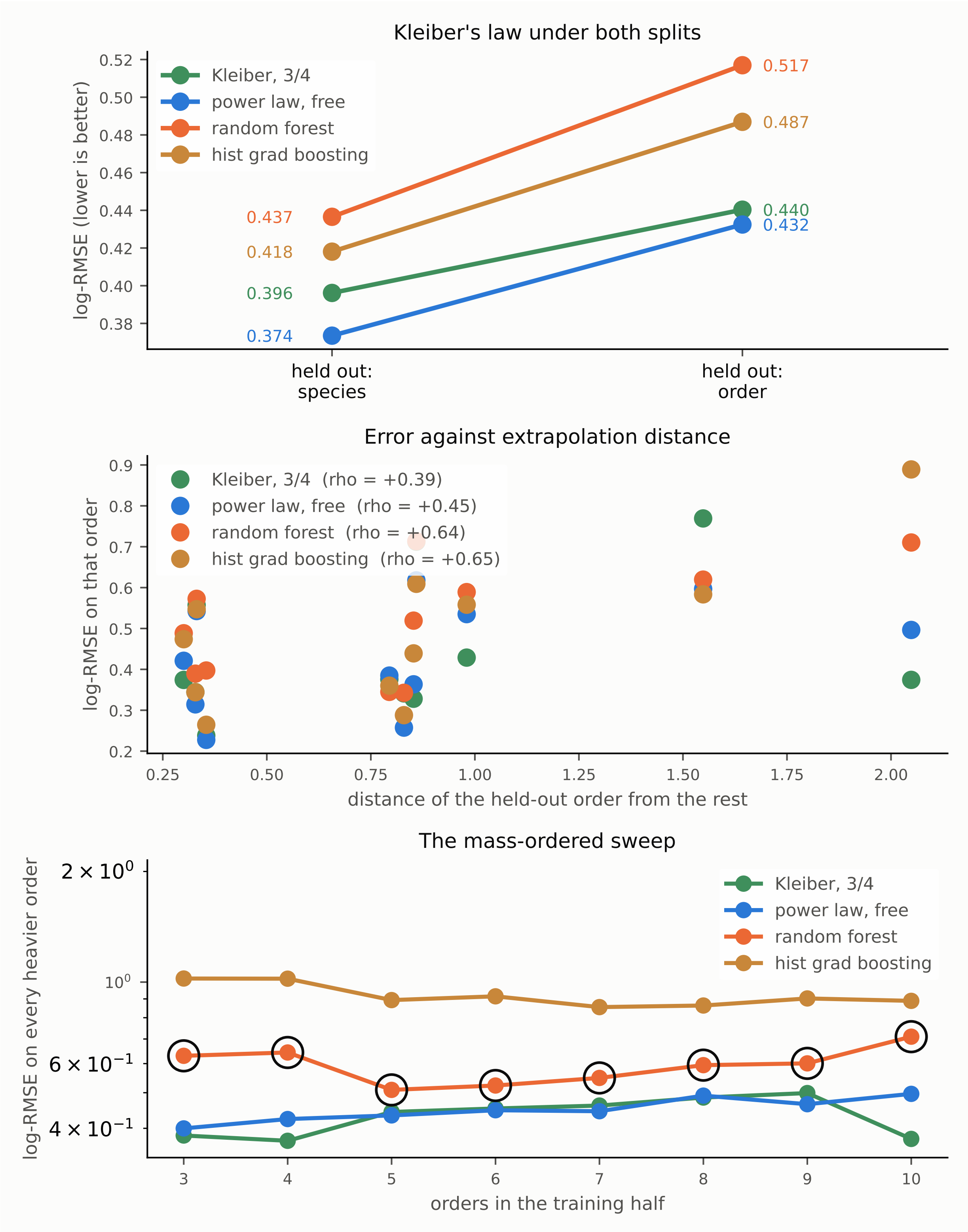

Figure 5: Kleiber's law under the same three splits. Top: no inversion, because the trees lose under both splits and so have no interpolation margin to give up. Middle: error against extrapolation distance, rising for the flexible models as it does on HDB5. Bottom: the mass-ordered sweep, with the circled points marking cuts where the random forest is held inside its training range.

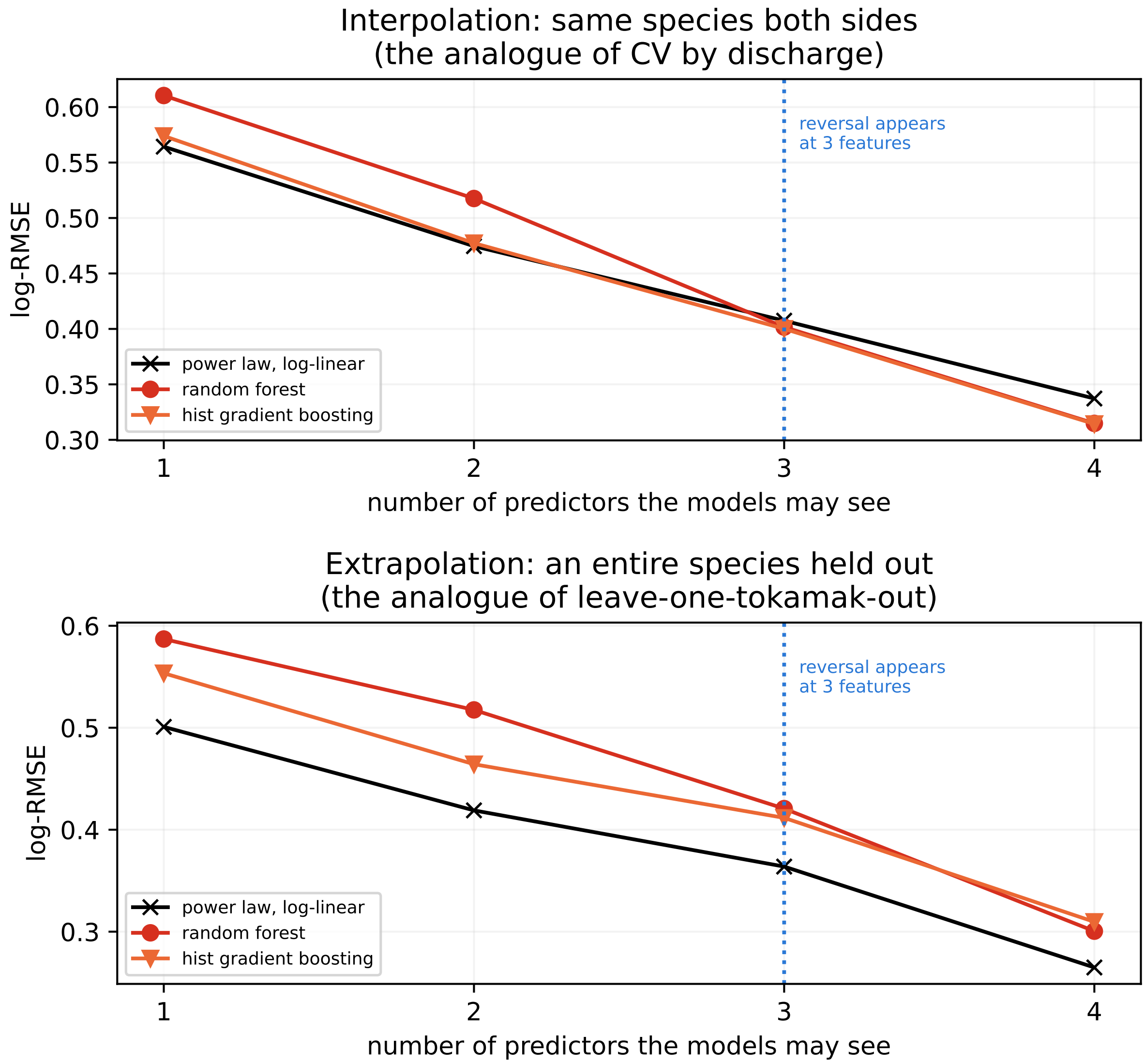

Figure 6: The reversal's precondition, on 3599 plants held fixed. Top: the interpolation split, where every species appears on both sides. Bottom: an entire species held out. Only the number of predictors varies. The top curves cross; the bottom ones do not.